

MECHANICS

Mechanics:

It is the science which deals with moving bodies or bodies at rest under the action of some forces.

Statics:

It is the branch of mechanics which deals with the action of forces on bodies, the forces being so arranged that the bodies are at rest.

Dynamics:

Dynamics is that branch of mechanics which deals with the motion of bodies under the action of given forces.

Kinematics:

Kinematics is the study of geometry of motion without to the cause of motion. It deals with displacement, velocity, acceleration etc. and we establish relations between these and time without reference to the cause of motion.

Kinetics:

Kinetics is the study of relationship between the forces and the resulting motion of bodies on which they act.

Topics in Statics:**Matter:**

Anything that occupies space and is perceived by our senses is matter. Table, cup air, water, etc. are example of matter.

Body:

A body is a portion of matter occupying finite space. It has, therefore, a definite volume and a definite mass.

Particle:

A particle is a body indefinitely small, so that the distance between its different parts is negligible. It may be regarded as a mathematical point associated with mass.

Rigid Body:

A body is said to be rigid when it does not change its shape and size when subject to external forces i. e., a rigid body is body in which the distance between any two points always remains the same.

Force:

Force is an agent which changes or tends to change the state of rest or uniform motion of a body. Force is a vector quantity.

Representation of a force:

A force is completely known if we know:

- (i) its magnitude
- (ii) its direction
- (iii) its point of application

Thus, we can completely represent a force by a straight line AB drawn through the point of application A along the line of action of the force, the length of the line AB representing the magnitude of the force and the order of the letters A, B specifying the direction.

Equilibrium of forces:

A system of forces acting on a body is said to be in equilibrium if it produces no change in the motion of the body.

Equilibrium of Two Forces:

Two forces acting at a point are in equilibrium if and only if they,

- (i) are equal in magnitude
- (ii) act along the same line
- (iii) have opposite directions

Forces in Statics:

1. Action and Reaction :

Whenever one body is in contact with another body, there act equal and opposite forces at the point of contact. Such forces are called action and reaction.

2. Weight:

Every body is attracted towards the centre of the earth with a force proportional to its mass (the quantity of matter in the body). This force is called the weight of the body. If m is the mass of the body and g , the acceleration due to gravity, then its weight $W = mg$.

Note:

- (i) Weight always act vertically downwards.
- (ii) Weight always act at the centre of gravity of a body.
- (iii) Centre of gravity of a smooth rod is its middle point.
- (iv) By 'a light body' we mean, a body without weight.

Particular Case:

- (a) If P and Q are at right angles to each other, i. e., $\alpha = 90^\circ$, then

$$R = \sqrt{P^2 + Q^2} \text{ and } \tan \theta = \frac{Q}{P}$$
- (b) If $P = Q$, then $R = 2P \cos \frac{\alpha}{2}$ and $\theta = \frac{\alpha}{2}$.
- (c) If P and Q are in the same direction, i. e., $\alpha = 0$, then R is the same direction as P and Q are $R = P + Q$. This is called the greatest resultant of the two forces.
- (d) If P and Q are in the opposite of P and $R = P - Q$. This is called the least resultant of the two forces.

Components of a Force in Two Given Directions:

The components of a force F along directions making angles α and β with it are

$$\frac{F \sin \beta}{\sin (\alpha + \beta)} \text{ and } \frac{F \sin \alpha}{\sin (\alpha + \beta)} \text{ respectively}$$

Resolved Parts of a Force:

When the components of a force are at right angles to each other they are called resolved parts of the force.

The resolved part of a force in a given direction is obtained multiplying the given force by the cosine of the angle between the line of action of the given force and the given direction i. e., the resolved part of a force F in the direction making an angle θ with the direction of the force is $F \cos \theta$.

Resultant of a Number of Coplanar and Concurrent Forces:

The resultant of a number of concurrent forces is obtained using following steps.

- Choose two mutually perpendicular directions OX and OY along which the forces are to be resolved.
- Find the algebraic sum of the resolved parts of the forces along these lines. Denotes these sums by X and Y respectively.
- The magnitude of the resultant R is given by

$$R = \sqrt{X^2 + Y^2}$$

- If the resultant R makes an angle α with OX , then

$$\tan \theta = \frac{Y}{X}$$

This gives the direction of the resultant force R .

Condition of Equilibrium of a Number of Coplanar Concurrent Forces:

A gives number of forces acting at a point are in equilibrium if and only if the algebraic sum of their resolved parts in each of the two perpendicular directions OX and OY vanish separately.

Triangle Law Forces:

If three forces, acting at a point, represented in magnitude and direction by the three sides of triangle, taken in order, then they are in equilibrium.

Remark:

Since, by triangle law,

$$\vec{AB} + \vec{BC} + \vec{CA} = 0$$

Therefore $\vec{AB} + \vec{BC} = -\vec{CA}$ or $\vec{AB} + \vec{BC} + \vec{CA} = 0$

Thus, the triangle law of forces can also be stated as:

If two forces acting at a point be represented in magnitude and direction by the two sides of a triangle, taken in order, their resultant is given in magnitude and direction by the third side of the triangle taken in the opposite order.

Converse of Triangle Law of Forces:

If three forces acting at a point are in equilibrium, they can be represented in magnitude and direction by the sides of any triangle drawn in such a way as to have its sides parallel to the forces taken in order.

Note:

If three coplanar forces acting upon a rigid body keep it in equilibrium, they are either parallel or concurrent (all meeting in a point).

Polygon Law of Forces:

If a number of forces, acting at a point, be represented in magnitude and direction by the sides of a polygon, taken in order, the forces shall be in equilibrium.

LAMI'S THEOREM:

If three forces acting at a point be in equilibrium, then each force is proportional to the angle between the other two.

Let P, Q and R be three forces and α , β , γ be the angles between Q, R; R, P and P, Q respectively. If the forces are in equilibrium, then we have

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

Converse of Lami's Theorem:

If three forces acting at a point be such that each is proportional to the sine of the angle between the other two, then the three forces are in equilibrium.

$\lambda - \mu$ Theorem:

The resultant of two forces, acting at a point O along OA and OB and represented in magnitude by λ . OA and μ , OB, is represented by a force $(\lambda + \mu)$. OC, where C is a point on AB such that $\lambda CA = \mu CB$, i. e., C divides AB in the ratio $\mu : \lambda$.

In vector notation the above statement can be written as:

$$\lambda \cdot \vec{OA} + \mu \cdot \vec{OB} = (\lambda + \mu) \cdot \vec{OC}$$

Where C is a point on AB dividing it in the ratio $\mu : \lambda$.

Like and Unlike Parallel Forces:

Two parallel forces are said to be like when they act in the same direction and unlike when they act in the opposite directions.

Note:

- (i) If three parallel forces acting on a rigid body are in equilibrium, each is proportional to the distance between the other two.
- (ii) The algebraic sum of the resolved parts of any two parallel forces (not forming a couple) along any direction is equal to the resolved part of their resultant along same direction.

Moments of a Force about a Point:

The moment of a force about a point is the measure of its turning effect about that point. It is equal to the force and the $\perp$ distance of the point from the line of action of the force.

If F be a force and p be the $\perp$ distance of its line of action from the fixed point O, then moment of F about O = Fp .

Sign of Moment:

The moment of a force F about at point O is +ve or -ve according as the force rotates the body about O in the anti-clockwise or clockwise direction.

Working Rule to Decide the Sign of Moment:

If we walk along the line of action in the direction of the and find that the point about which we taking moments falls to out left, then the moment of the force about the point will be + ve and if it falls to the right, then the moment is negative.

Note:

- (i) If the moment of a force about a point is zero, then either the force is zero, or it passes through the point, i. e., $Fp = 0$
 $\Rightarrow$ Either $F = 0$ or $p = 0$.
- (ii) The moment of a force about a point on its line of action is zero, because in such a case $p = 0$.

Varignon's Theorem of Moments:

The algebraic sum of moments of any number of coplanar forces about any point in their plane is equal to the moment of their resultant about the same point.

Couple:

Two equal and unlike parallel forces with different lines of action are said to form a couple.

Arm of the Couple:

The $\perp$ distance between the lines of action of the forces forming the couple is known as the arm of the couple.

Note:

A couple, each of whose forces is P and the length of whose arm is p , is usually denoted by (P, p) .

Moment of a Couple:

The moment of a couple is obtained in magnitude by multiplying the magnitude of one of the forces forming the couple and the are m of the couple.

Thus, the moment of the couple (P, p) is P, p , i. e., Moment of a couple = Force $\times$ Arm of the couple.

Note:

Moment of a couple can never be zero.

Sign of the Moment of a Couple:

The moment of a couple is regarded as positive or negative according as it has a tendency to turn the body in the anti-clockwise or clockwise direction.

Important Results:

- (i) The algebraic sum of the moments of the two forces forming a couple about point in their plane is constant and equal to the moment of the couple.
- (ii) The moment decides the tendency to produce rotation. The greater moment, the greater is the tendency for rotatory motion.
- (iii) The sum of the resolved parts of the forces forming a coupl in any direction is zero.
- (iv) The resultant of a number of coplanar couples is equivalent to single couple whose moment is equal to the algebraic sum of the moments of the given couples.

- (v) If three forces acting on a rigid body be represented in magnitude, direction and line of action by the sides of a triangle, taken in order, then they are equivalent to a couple whose moment is represented by the twice the area of the triangle.

Friction:

The property by virtue of which a resisting force is created between two rough bodies which prevents the sliding of one body over the other is called the friction and this force which always acts in the direction opposite to that in which the body has a tendency to slide or move is called force of friction.

Limiting Friction:

When one body is just on the point of sliding on another body, the force of friction called into play attains its maximum value and is called limiting friction and the equilibrium then is said to be limiting equilibrium.

Static Friction:

When a body in contact with another body is in any position of equilibrium but not equilibrium, then the friction exerted is called static friction. Thus, static friction is less than the limiting friction.

Dynamic Friction:

When motion ensues by one body sliding on the other, the friction exerted between the bodies is called dynamic friction.

Sliding Friction:

If the body is sliding, the force of friction that comes into play is called sliding friction.

Rolling Friction:

If the body is rolling, the force of friction that comes into play is called rolling friction.

Laws of Friction:

The following laws govern the different of friction i.e., static, limiting and dynamic friction.

Law of Static Friction:

- (i) The direction of friction is opposite to the direction in which the body tends to move.
- (ii) The magnitude of the force of friction is just sufficient to prevent the body from moving.

Laws of Limiting Friction:

- (i) Limiting friction is equal in magnitude and opposite in direction to the force which tends to produce motion.
- (ii) The magnitude of limiting friction at the point of contact between two bodies bears a constant ratio to the normal reaction at the point.
- (iii) The constant ratio depends entirely on the nature of the material of which the surfaces in contact are composed of and is independent of their extent and shape.

Laws of Dynamic Friction:

- (i) The direction of dynamic friction is opposite to that in which the body is moving.
- (ii) The magnitude of dynamic friction bears a constant ratio to the normal ratio on the body but ratio is slightly less than the coefficient of friction in the case of limiting friction.

- (iii) The dynamic friction is independent of the velocity of motion.

Coefficient of Friction:

When a rough body is on the verge of sliding on another, the friction exerted bears a constant ratio to the normal reaction. This ratio of the limiting friction to the normal reaction is called the coefficient of friction. It is usually denoted by μ .

Angle of Friction:

When one body, placed on another body is in limiting equilibrium, the friction exerted is the limiting friction. In this case, the angle which the resultant of the force of friction and the normal reaction makes with the normal reaction at the point of contact is called the angle of friction is usually denoted by λ .

Note: The coefficient of friction is equal to the tangent of the angle of friction.

Cone of Friction:

The right cone described with its vertex at the point of contact of two rough bodies and having the common normal at the point of contact as axis and the angle of friction as the semi-vertical angle, is called the cone of friction.

Least Force on Horizontal Plane:

The least force required to pull a body of weight W on the rough horizontal plane is $W \sin \lambda$.

Least Force on Inclined Plane:

Let α be the inclination of rough inclined plane to horizontal and λ , the angle of friction.

- (i) If $\alpha = \lambda$, then the body is in limiting equilibrium and is just on the point of moving downwards.
- (ii) If $\alpha < \lambda$, then the least force required to pull a body of weight W down the plane is $W \sin (\lambda - \alpha)$.
- (iii) If $\alpha > \lambda$, then the body cannot rest on the plane under its own weight and reaction of the plane. So, the question of finding the least force does not arise.

Note:

The least force required to pull a body of weight W up an inclined rough plane is $W \sin (\alpha + \lambda)$.

Centre of Gravity:

The centre of gravity of a body or system of particles rigidly connected together, is that point through which the line of action of the weight of the body always passes.

Position of Centre of Gravity in Some Special Cases:

- (i) *Uniform Rod:* At its mid point.
- (ii) *Parallelogram, Rectangle or Square:* At the intersection of the diagonals.
- (iii) *Triangular lamina:* At a distance from the centre on the symmetrical radius.
- (iv) *Circular Arc:* At distance $\frac{(a \sin \alpha)}{\alpha}$ from the centre on the symmetrical radius.

Where a = radius and 2α = angle subtended by the arc at the centre.

- (v) *Sector of a Circle:* At a distance $\frac{2a}{3} \cdot \frac{\sin \alpha}{\alpha}$ from the centre on the symmetrical radius.
- (vi) *Semi-Circular Arc:* At a distance $\frac{4\alpha}{3\pi}$ from the centre on the symmetrical radius, where α is the radius.
- (vii) *Hemisphere:* At distance $\frac{3a}{8}$ from the centre on the symmetrical radius, where a is the radius.
- (viii) *Hemispherical Shell:* At a distance $\frac{a}{2}$ from the centre on the symmetrical radius, where a is the radius.
- (ix) *Solid Cone:* At a distance $\frac{h}{4}$ from the base on the axis, where h is the height of the cone.
- (x) *Conical Shell:* At a distance $\frac{h}{3}$ from the base on the axis, where h is the height of the cone.

Topics in Dynamics:

Rest and Motion:

A particle is said to be at rest if it does not change its position with respect to its surroundings and is said to be in motion if it changes its position with respect to its surroundings.

Path:

The straight line or the curve along which an object moves is called its path.

If the path is straight line, the object is said to have **rectilinear motion** and if the path is curve (in plane or in space), the object is said to have **curvilinear motion**.

Speed:

The speed of a moving point is the rate at which it describes its path. Speed is a scalar quantity.

Average Speed:

Average speed of a moving object in a time interval is defined as the distance travelled by the object during that time interval divided by the time interval.

Thus,

$$\text{average speed in a time interval} = \frac{\text{distance travelled in the given time interval}}{\text{time interval}}$$

Displacement:

The displacement of moving points is the distance covered by it in a definite direction.

A displacement has two fundamental characteristics-magnitude and direction. So, it is a vector quantity.

Uniform Velocity:

A particle is said to move with uniform velocity if it moves in a constant direction and covers equal distances in equal intervals of time, however small these intervals may be.

Constant Velocity:

A particle is said to move with constant velocity if it covers distance in equal intervals of time, even if it is moving in a constant direction.

Average Velocity:

The average velocity of a particle in a given interval of time is given by the ratio of the total displacement undergone to the total time taken.

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time}}$$

Acceleration and Retardation:

The acceleration of a particle is the rate of increase in its velocity while retardation is the rate of decrease in velocity. Thus, retardation is negative acceleration.

Units of acceleration are cm/sec^2 and m/sec^2 .

Acceleration has magnitude as well as direction. So, it is a vector quantity.

Uniform Acceleration:

A particle is said to moving with uniform acceleration, if equal changes in velocity take place in equal intervals of time, however small these intervals may be.

Equations of Motion:

If a particle moves along straight line with initial velocity u and constant acceleration f , then the following relations are known as equations of motion:

$$(i) \quad v = u + ft \qquad (ii) \quad s = ut + \frac{1}{2}ft^2 \qquad (iii) \quad v^2 - u^2 = 2fs$$

Where 'v' is the velocity after time t and s , the distance travelled in this time.

Motion Under Gravity:

When a body is allowed to fall towards the earth, it will move vertically downwards with an acceleration which is always the same the same place on the earth but varies slightly from place to place. This acceleration is called *acceleration due to gravity*. Its value in F. P. S. system is 32 ft/sec^2 , in C. G. S. system is 981 cms/sec^2 and in M. K. S. system is 9.8 m/sec^2 . It is denoted by g .

The acceleration due to gravity always acts vertically downwards. If the body moves downwards, then the effect of acceleration due gravity is to increase its velocity. If the body moves upwards, then the effect of acceleration due to gravity is that of retardation, i. e., the velocity of the body decreases. Hence, g is taken positive for the downwards motion and negative for the upward of it's a body.

Downward Motion:

If a body is projected vertically downwards from a point at a height h above the earth's surface with velocity u , the equation of its motion are

$$(i) \quad v = u + gt \qquad (ii) \quad h = ut + \frac{1}{2}gt^2$$

$$(iii) \quad v^2 = u^2 + 2gh \qquad (iv) \quad h_n = u + \frac{1}{2}g(2n - 1)$$

Where h_n denotes the distance covered in the n th second.

Upward Motion:

When a body is projected vertically upwards with initial velocity u , then it moves in a straight line with constant retardation g .

So, the equations of motion in this case are:

(i) $v = u - gt$

(ii) $h = ut - \frac{1}{2}gt^2$

(iii) $v^2 = u^2 - 2gh$

(iv) $h_n = u - \frac{1}{2}g(2n - 1)$

Newton's Laws of Motion:**First Law of Motion:**

Everybody continues in its of rest or of uniform motion in a straight line unless it is compelled by an external force (called impressed force) to change that state.

This law asserts that a force is necessary to change

- (a) the velocity of a body
- (b) the velocity of a body
- (c) the direction of motion

Second Law of Motion:

The rate of change of momentum is directly proportional to the impressed force and it takes place in the direction in which the force acts.

Third Law of Motion:

To every action, there is an equal and opposite reaction. This law asserts that forces occur in pairs. When a book is placed on the table, the book presses the table with a certain force (which is action) and the table in turn presses the book with an equal but opposite force (which is reaction).

Inertia:

A body has no tendency of itself to change its state of rest or motion if it is kept free from the action of external forces. The inability of a body to change its position by itself is called Inertia.

Newton's first law of motion is also, therefore, called the Law of Inertia.

Momentum of a Body:

It is the quantity of a motion possessed by a body and is equal to the product of its mass and velocity with which it moves.

Thus, if m be the mass of a body moving with velocity v , then its momentum is mv .

Momentum is vector quantity. The direction of the momentum is the same as that of velocity.

The units of momentum are kg m/sec or gm cm/sec.

Relation between Force, Mass and Acceleration:

If a force P acting on a body of mass m sets it in motion under acceleration f , the force P is given by $P = mf$. That is, Force causing motion in absolute units = mass $\times$ acceleration. This is called the fundamental equation of dynamics.

Note: The equation $P = mf$ is valid only when the mass of the body in motion remains constant whereas the force may be constant or variable. If the force is constant, the acceleration is uniform.

Units of Force:

Absolute Units:

- (a) Dyne: In CGS system, the unit of force is dyn. 1 dyne is the force which produces an acceleration of 1 cm/sec^2 in a mass of 1g.
- (b) Newton: In MKS system, the unit of force is Newton. 1N is the force which produces an acceleration of 1 m/sec^2 in a mass of 1 kg.
- (c) Poundal: In FPS system, the unit of force is Poundal. 1 Poundal is the force which produces an acceleration of 1 ft/sec^2 in a mass of one pound.

Note: These unit of force are called absolute units because their values are the same everywhere and do not upon the value of 'g' which varies from place to place on the Earth's surface.

Gravitational Units:

The units of force which depend on the value of 'g' are called gravitational units of force.

- (a) In CGS system, 1 gm. Wt. = g dynes = 981 dynes
- (b) In KMS system, 1 kg. wt. = g Newtons = 9.8 N.
- (c) In FPS system, 1 lb. wt. = g Poundal = 32 poundals.

Note: While using the formula $P = mf$, P is always measured in absolute units i. e., in poundals or dynes or Newtons.

Projectiles:

A particle projected in any direction is called a projectile.

Trajectory:

The path described by the particle is called trajectory.

Velocity of Projection:

The velocity with which the particle is projected is called the velocity of projection.

Angle of Projection:

It is the angle which the direction of projection makes with the horizontal.

Range:

The distance between the point of projection and the point where the projectile hits a given plane through the point of projection is called its range.

When the given plane is horizontal, it is called *horizontal range*.

Time of Flight:

The time taken between the instant of projection and the instant when the projectile meets a fixed plane through the point of projection is called the time of flight.

Greatest Height:

The maximum height reached by the particle above the point of projection during its motion is called the greatest height.

Multiple Choice Questions:

1. The branch of science which deals with action of forces on bodies is
 - (a) Mechanics
 - (b) Statics
 - (c) Dynamics
 - (d) Physics
2. The branch of mechanics which deals with bodies at motion is
 - (a) Statics
 - (b) Dynamics
 - (c) Kinematics
 - (d) None
3. The branch of mechanics which deals with bodies at rest is
 - (a) Statics
 - (b) dynamics
 - (c) Kinematics
 - (d) None
4. The branch of dynamics which deals with geometry of motion of bodies, without taking into account their masses is
 - (a) Kinetics
 - (b) Kinematics
 - (c) Statics
 - (d) None
5. Which of the following branch of dynamics is concerned with the relations between the forces acting on bodies and the resulting motion.
 - (a) Kinematics
 - (b) Kinetics
 - (c) Statics
 - (d) None
6. Which of the following is equation of motion.
 - (a) $\vec{P} = m \vec{v}$
 - (b) $\vec{F} = m \vec{a}$
 - (c) $\vec{F} = \frac{GMm}{r^2}$
 - (d) None
7. Forces whose lines of action intersect at one point are called
 - (a) colinear forces
 - (b) coplanar forces
 - (c) concurrent forces
 - (d) unit forces
8. If a particle is in equilibrium under the action of several forces, then these forces may be represented by the side of a
 - (a) triangle
 - (b) rectangle
 - (c) pentangle
 - (d) polygon
9. If a particle is in equilibrium under the action of three forces then these forces may be represented by the sides of a
 - (a) triangle
 - (b) rectangle
 - (c) parallelogram
 - (d) polygon
10. If a particle is in equilibrium under the action of three forces then each force has a magnitude proportional to the _____ of angle between the other two.
 - (a) tangent
 - (b) secant
 - (c) sine
 - (d) cosine
11. Let three forces act perpendicularly to the sides of a triangle at their middle points and are proportional to the sides. Then forces are
 - (a) colinear
 - (b) inequilibrium
 - (c) constant
 - (d) none
12. The tendency of a force to turn a body about a point is called
 - (a) couple
 - (b) resultant
 - (c) moment
 - (d) magnitude

13. The moment about a point O of the resultant of a system of concurrent forces is equal to the _____ of the moments of the various forces about the same point O.
(a) magnitude (b) sum
(c) difference (d) product
14. The sum of the moments of the two forces of a couple about any given point is
(a) zero (b) negative
(c) positive (d) not zero
15. A couple can be transferred in its
(a) circle (b) line of action
(c) plane (d) none
16. The system of coplanar couples with moments G_1, G_2, G_3 are in equilibrium if:
(a) $G_1 G_2 G_3 = 0$ (b) $G_1 - G_2 - G_3 = 0$
(c) $G_1 + G_2 + G_3 = 0$ (d) $G_1 + G_2 = G_3$
17. If three forces are represented by the sides of a triangle taken in order then they are equivalent to a
(a) single force (b) couple
(c) resultant of forces (d) none
18. If three forces act perpendicularly to the sides of a triangle at their middle points and are proportional to the sides, then forces are
(a) coplanar (b) in equilibrium
(c) constant (d) none
19. Let A be a point on the line of action of a force $\vec{F}$ and $\vec{r}$ be a position vector of A with respect to the fixed reference point O. Then moment of force F about O is:
(a) $\vec{M} = \vec{r} \cdot \vec{F}$ (b) $\vec{M} = \vec{r} \times \vec{F}$
(c) $\vec{M} = \vec{r} + \vec{F}$ (d) $\vec{M} = \vec{r} \vec{F}$
20. The force derive from a potential energy function through the relation $\vec{F} = -\nabla V$ is independent of:
(a) origin (b) coordinates
(c) time (d) none
21. If the only force acting on a projectile is the gravity, then the equation of motion is:
(a) $\frac{d^2 \vec{r}}{dt^2} = \vec{g}$ (b) $\vec{F} = m \vec{a}$
(c) $\frac{d \vec{r}}{dt} = m \vec{g}$ (d) $\vec{F} = \vec{a} \vec{g}$
22. A particle moving in a bounded region is perform:
(a) free motion (b) bounded motion
(c) unbounded motion (d) harmonic motion

23. The elating string can only be extended by applying a
(a) load (b) tension (c) tensile force (d) all of these
24. A string is called linear string if it obeys:
(a) Newton's law (b) Kepler's law
(c) Hooke's law (d) Fermet's law
25. Which of the following force is conservative:
(a) $\vec{F} = (2xy + z^3)\hat{i} + (x^2 + 2y)\hat{j} + (3xz^2 - 2)\hat{k}$
(b) $\vec{F} = 2xyz\hat{i} + x^2z\hat{j} + x^2y\hat{k}$
(c) $\vec{F} = 2xz\hat{i} + (x^2 - y)\hat{j} + (2z - x^2)\hat{k}$
(d) all of these
26. A particle moves along the x-axis have point of stable equilibrium $x = 0$ in a force field having potential $V =$
(a) $\frac{kx^2}{2}$, $k > 0$ (b) $x^2(6 - x)$ (c) x^4 (d) All of these
27. A particle is moving in a force field with potential $V = x^2 + y^2 + z^2 - 8x + 16y - 4z$. Then which of the following is a point of stable equilibrium.
(a) 2 (b) 4 (c) -8 (d) all of these
28. The general motion of a rigid is:
(a) translation (b) rotation
(c) screw motion (d) all of these
29. If a particle is moving in a plane with constant speed, then its acceleration is _____ to its velocity.
(a) equal (b) parallel
(c) perpendicular (d) proportional
30. The angular velocity of earth is:
(a) 8.14×10^{-3} (b) 7.29×10^{-5}
(c) 6.12×10^{-4} (d) none
31. The role of moment of inertia in rotational motion is same as the role of _____ in linear motion.
(a) work done (b) force
(c) tension (d) mass
32. The 3×3 inertia matrix is also a Cartesian tensor of rank:
(a) 1 (b) 2 (c) 3 (d) 4
33. The moment of inertia of a ring of radius 'a' about an axis through its centre is:
(a) $\frac{1}{3} Ma^2$ (b) $\frac{1}{2} Ma^2$
(c) Ma^2 (d) none
34. The moment of inertia of a uniform elliptical plate with semi-major axis 'a' and semi-minor axis 'b' about a major axis is:
(a) $\frac{1}{3} Ma^2$ (b) $\frac{1}{4} Mb^2$
(c) $\frac{1}{2} Mb^2$ (d) Ma^2

35. The moment of inertia of a uniform triangular lamina of mass M about one of its sides is
(a) $\frac{1}{3} Mh^2$ (b) $\frac{1}{4} Mh^2$
(c) $\frac{1}{5} Mh^2$ (d) $\frac{1}{6} Mh^2$
36. The moment of inertia of a right circular cone about its axis of symmetry:
(a) $\frac{1}{3} Ma^2$ (b) $\frac{3}{5} Ma^2$
(c) $\frac{3}{10} Ma^2$ (d) $\frac{1}{12} Ma^2$
37. The moments of inertia about all lines through the centre of mass of a uniform square lamina and lying in its plane are:
(a) equal (b) not equal
(c) different (d) parallel
38. The centrifugal force acting on a train of mass 100,000 kg at the equator is:
(a) 343 N (b) 330 N
(c) 650 N (d) none
39. The centrifugal force acting on a train of mass 100,000 kg at the point with latitude 60° is:
(a) 343 N (b) 171.5 N
(c) 190 N (d) 200 N
40. If all the points of a rigid body move parallel to a given fixed plane, then the motion is called:
(a) centrifugal motion (b) coriolis motion
(c) harmonic motion (d) plane motion
41. If plane motion is rotation about a fixed axis, then system has _____ degree of freedom.
(a) 1 (b) 2 (c) 3 (d) 4
42. In general plane motion the degree of freedom of system is:
(a) 1 (b) 2 (c) 3 (d) 4
43. The axis about which the body rotates is called:
(a) central axis (b) instantaneous axis of rotation
(c) symmetric axis (d) none
44. The coefficient of friction is usually denoted by:
(a) Δ (b) δ (c) μ (d) ε
45. If a solid cylinder of radius a and mass M rolls without slipping down an inclined plane of angle α , then the acceleration is:
(a) 0 (b) 150° (c) $\frac{1}{2} g \sin \alpha$ (d) constant
46. When the body is rotating about an instantaneous axis through the fixed point, its angular velocity can be expressed in terms of the
(a) space variable (b) time variable
(c) time derivatives (d) central force
47. The plane containing the orbit of earth's motion around the sun is called:
(a) steroid (b) circle
(c) ecliptic (d) ring
48. The equator is inclined to the ecliptic about:
(a) 25° (b) 19° (c) 23.5° (d) 91.23°

49. Which of the following has greatest effect on earth.
(a) sun (b) moon (c) stars (d) satellites
50. The axes relative to the products of inertia are zero are called:
(a) inertia of axes (b) principal of axes
(c) altitude of axes (d) none
51. Number of principal axes of a rigid body is:
(a) 2 (b) 3 (c) 4 (d) 5
52. The principal moments of inertia are always:
(a) integer (b) rational
(c) real (d) complex
53. A rigid body is called spherical top if all three principal moments are:
(a) distinct (b) real
(c) orthogonal (d) equal
54. If all the three principal moments at the centroid are distinct, then the rigid body is called:
(a) asymmetrical top (b) spherical top
(c) symmetrical top (d) orthogonal top
55. If two distributions have same moment of inertia then they are called:
(a) orthogonal (b) symmetrical
(c) equimomental (d) isomorphic
56. If two systems are equimomental then which of following is true.
(a) they have same mass (b) they have same centroid
(c) they have same principal axes (d) all of these
57. Which of the following is equimomental of hoop of mass m and radius $\frac{a}{\sqrt{2}}$.
(a) circular plate of mass m and radius a (b) triangular plate of mass m
(c) square lamina of side length a (d) none
58. If F is frictional force, N is normal reaction and μ is coefficient of friction then which of the following is law of static friction.
(a) $F = \mu N$ (b) $F \leq \mu + N$
(c) $F \leq \mu N$ (d) $F \geq \mu N$
59. The total number of independent coordinates for a given mechanical system is called:
(a) degree (b) degree of freedom
(c) order (d) length of system
60. 5.97×10^{24} kg is a mass of:
(a) sun (b) earth (c) moon (d) star
61. Mass of sun is:
(a) 1.987×10^{30} kg (b) 5.97×10^{24} kg
(c) 3.84×10^8 kg (d) 7.34×10^{22} kg
62. 1.74×10^6 metre is radius of:
(a) sun (b) earth (c) moon (d) non
63. 9.7805 m/sec^2 is gravity on the
(a) earth (b) poles
(c) equator (d) sun
64. 6371 km is radius of:
(a) earth (b) sun

65. 0.511 MeV/c² is:
- | | |
|-------------------|------------------|
| (c) moon | (d) poles |
| (a) proton mass | (b) neutron mass |
| (c) electron mass | (d) none |

KEY to MCQ of Mechanics

1	a	2	b	3	a	4	b	5	b	6	b
7	c	8	d	9	a	10	c	11	b	12	c
13	b	14	d	15	c	16	c	17	b	18	b
19	b	20	c	21	a	22	b	23	c	24	d
25	d	26	d	27	d	28	c	29	b	30	d
31	c	32	c	33	b	34	d	35	c	36	a
37	a	38	b	39	d	40	a	41	c	42	b
43	c	44	d	45	c	46	c	47	c	48	c
49	b	50	b	51	b	52	c	53	d	54	a
55	c	56	d	57	a	58	c	59	c	60	b
61	a	62	c	63	c	64	a	65	c		